INDIAN INSTITUTE OF INFORMATION TECHNOLOGY RAICHUR

MATHEMATICS CLUB

PROBLEM NUMBER: WP2602

DATE: JANUARY 18, 2026

DUE DATE: FEBRUARY 3, 2026

WEEKLY PROBLEM

EXERCISE. **Spread of a Rumor Between Two Groups**

There are two groups of people in a college: **Group A** and **Group B**.
A rumor spreads continuously between the two groups over time.
Let $x(t)$ be the number of people in **Group A** who know the rumor at time t,
and $y(t)$ be the number of people in **Group B** who know the rumor at time t.
The spread follows these rules:

- **Group A** shares the rumor with **Group B**, and vice versa.
- People in both groups forget the rumor over time.

This process is modeled by the system:

$$\begin{aligned}\frac{dx}{dt} &= -x + y \\ \frac{dy}{dt} &= x - y\end{aligned}$$

In matrix form, define: $v(t) = [x(t), y(t)]^T$

Then: $$\frac{dv}{dt} = Av(t)$$

where $$A = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$$

Initially, at time $t = 0 : x(0) = 100$ and $y(0) = 0$, $I \implies$ *Identity Matrix*

(A) By thinking in terms of very small time steps, explain why the state after time t can be approximated by:

$$v(t) \approx (I + (t/n)A)^n v(0)$$

, for large n.

(B) Using limit definition of e^x:

$$e^x = \lim_{n\to\infty} \left(1 + \frac{x}{n}\right)^n$$

Prove:

$$e^{tA} = \lim_{n\to\infty} \left(I + \frac{t}{n}A\right)^n$$

(C) Without solving the system explicitly, explain why the total number of people who know the rumor remains constant, and why both groups eventually know the rumor equally.

(Scan this to submit solution!)

Top Submissions for last Week's Problem (WP2601):

1. **Aditya Chauhan** - CS25B1004
2. **Ashutosh Kumar** - AD23B1007